

Assignment 3: Block-wise SVD Image Compression

MATH 485 — Spring 2026

Implementation Summary

1. **Preprocessing:** Load image, convert to grayscale, center-crop so both dimensions are divisible by the block size.
2. **Block-wise SVD:** Partition the image into non-overlapping 8×8 blocks. For each block, compute the SVD and reconstruct using only the top- k singular values.
3. **Metrics:** For each $k \in \{1, \dots, 8\}$, compute:
 - Compression Ratio = $\frac{n^2}{k(2n+1)}$ where n is the block size
 - Reconstruction Error (Frobenius norm): $\|A - \hat{A}_k\|_F$
 - PSNR (Peak Signal-to-Noise Ratio)
4. **Extended analysis** with 4×4 and 16×16 block sizes.

Key code chunk:

```
# Block-wise SVD functions
compress_block <- function(block, k) {
  s <- svd(block)
  # Reconstruct with top-k singular values
  s$u[, 1:k, drop = FALSE] %*% diag(s$d[1:k], nrow = k) %*% t(s$v[, 1:k, drop = FALSE])
}

blockwise_svd <- function(A, block_size, k) {
  recon <- matrix(0, nrow(A), ncol(A))
  for (i in seq(1, nrow(A), by = block_size)) {
    for (j in seq(1, ncol(A), by = block_size)) {
      rows <- i:(i + block_size - 1)
      cols <- j:(j + block_size - 1)
      recon[rows, cols] <- compress_block(A[rows, cols], k)
    }
  }
  recon
}

# Metric functions
comp_ratio <- function(block_size, k) {
  block_size^2 / (k * (2 * block_size + 1))
}
```

Compression Analysis

k	Compression Ratio	Frobenius Error	PSNR (dB)
1	3.76×	2475.9	30.25
2	1.88×	1413.3	35.12
3	1.25×	786.8	40.21
4	0.94×	440.4	45.25
5	0.75×	221.6	51.21
6	0.63×	86.3	59.41
7	0.54×	18.9	72.60
8	0.47×	0.0	295.22

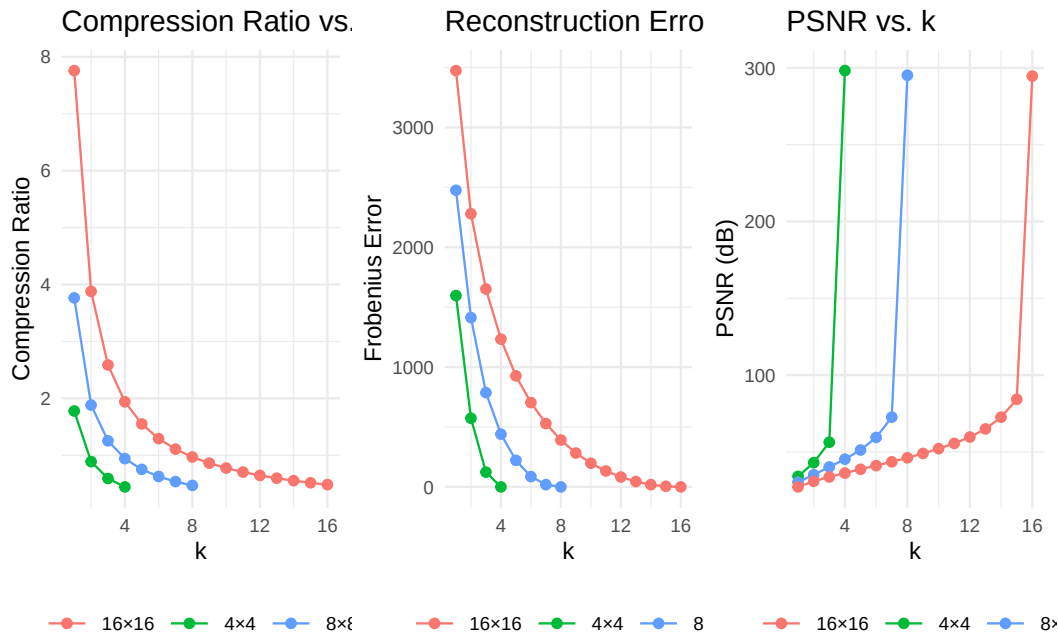
Only $k = 1$ ($3.76\times$) and $k = 2$ ($1.88\times$) achieve meaningful compression. At $k = 3$ the ratio drops to $1.25\times$. For $k \geq 4$, the retained SVD components exceed the original 64 values per block — no compression benefit.

Reconstruction quality improves rapidly: PSNR rises from ~ 30 dB at $k = 1$ to ~ 40 dB at $k = 3$ to ~ 51 dB at $k = 5$. By $k = 7$ the Frobenius error is near-zero.

Reconstructed Images

Metric Plots

Block Size Comparison



Larger blocks (16×16) achieve higher compression ratios at the same k , but introduce more visible blocking artifacts. Smaller blocks (4×4) compress less — fewer singular values to truncate — but preserve local detail better.



Figure 1: Reconstructions for $k=1$ through $k=8$ (8×8 blocks)

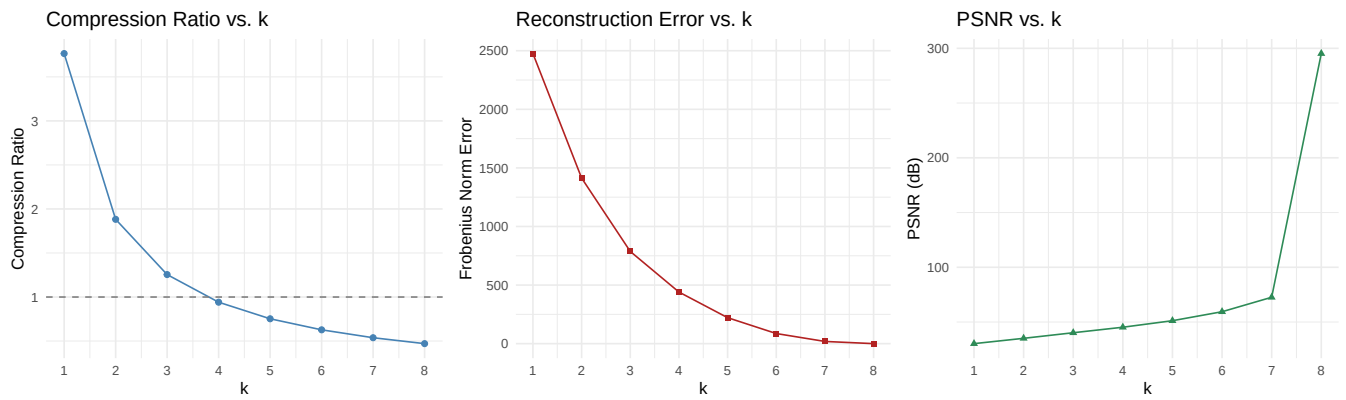


Figure 2: Compression ratio, reconstruction error, and PSNR vs. k